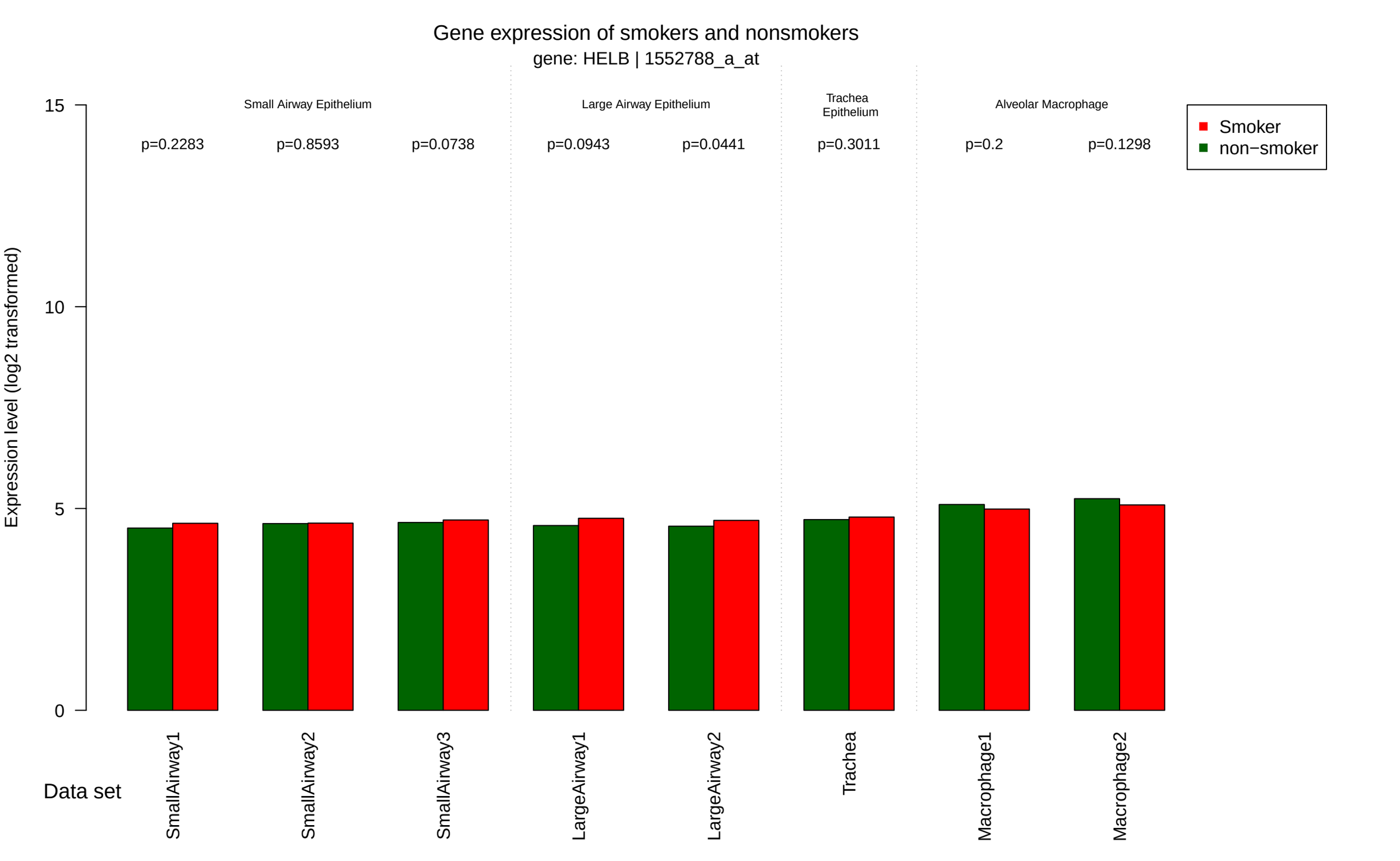
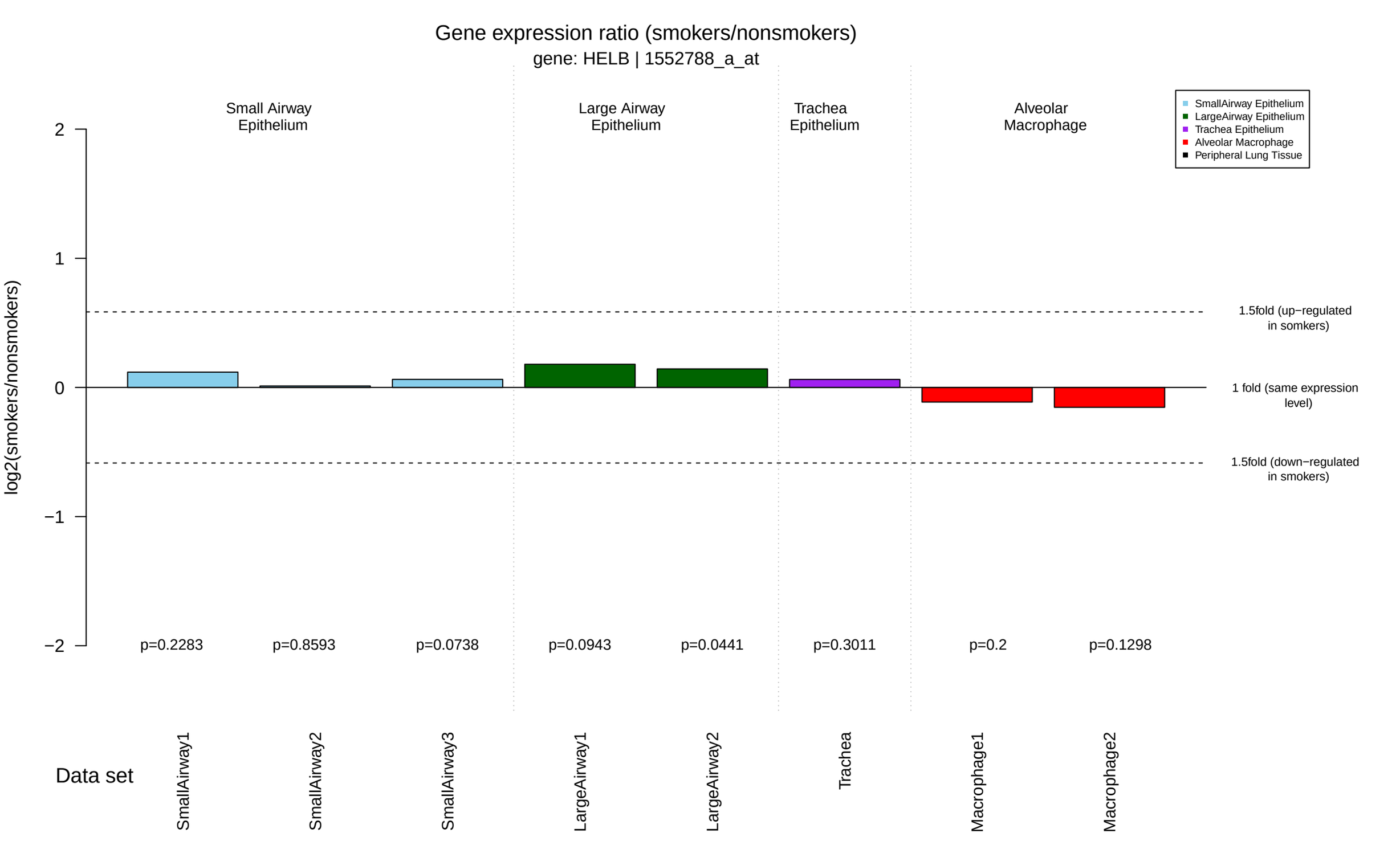
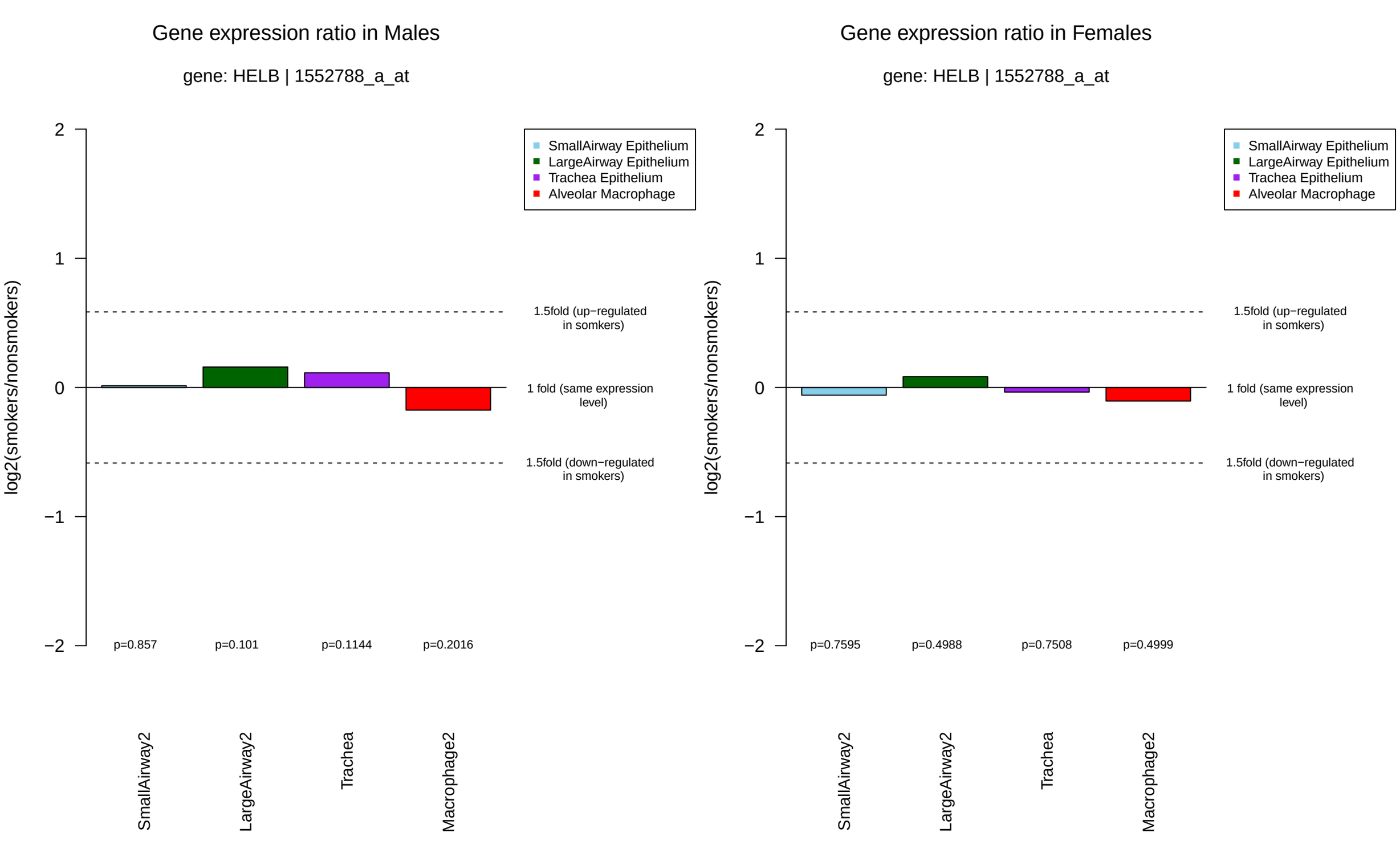
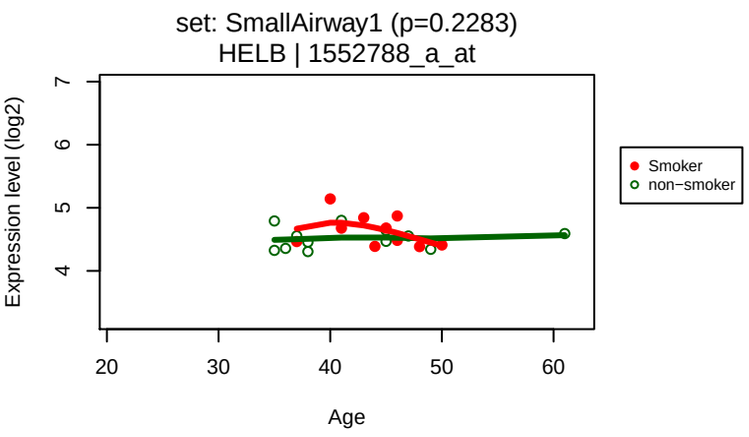
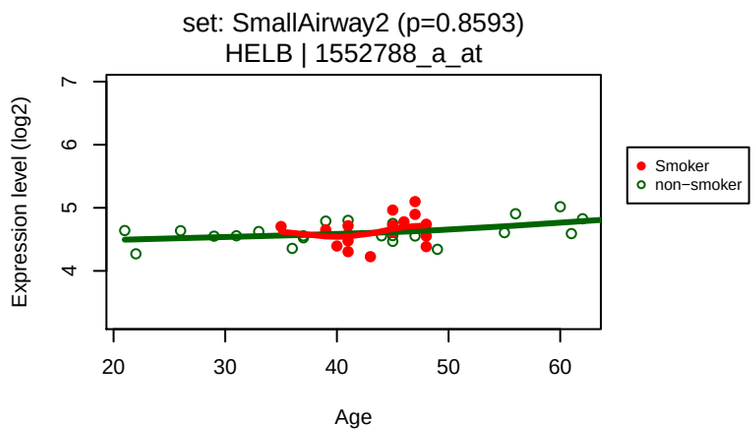
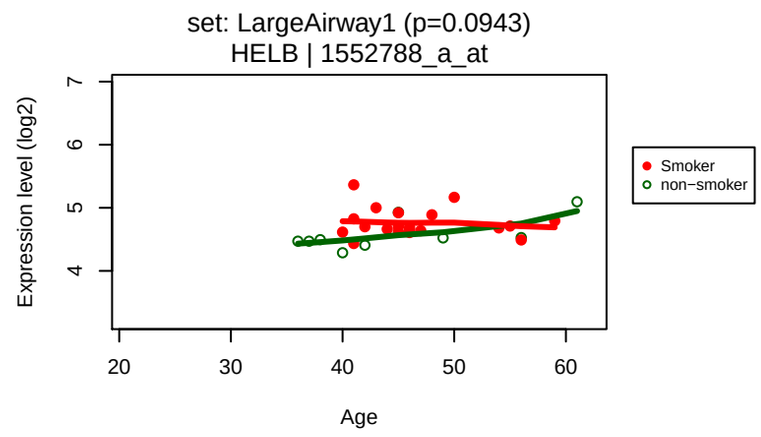

Gene expression of smokers and nonsmokers
gene: HELB | 1552788_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2283
p=0.8593
p=0.0738
p=0.0943
p=0.0441
p=0.3011
p=0.2
p=0.1298
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

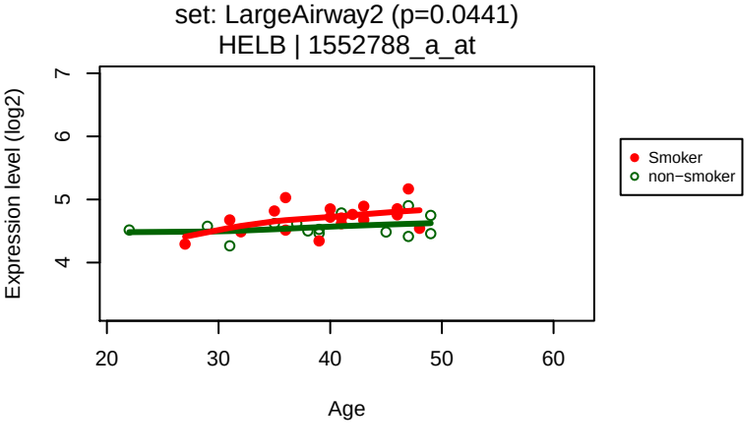

Gene expression ratio (smokers/nonsmokers)
gene: HELB | 1552788_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2283
p=0.8593
p=0.0738
p=0.0943
p=0.0441
p=0.3011
p=0.2
p=0.1298
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

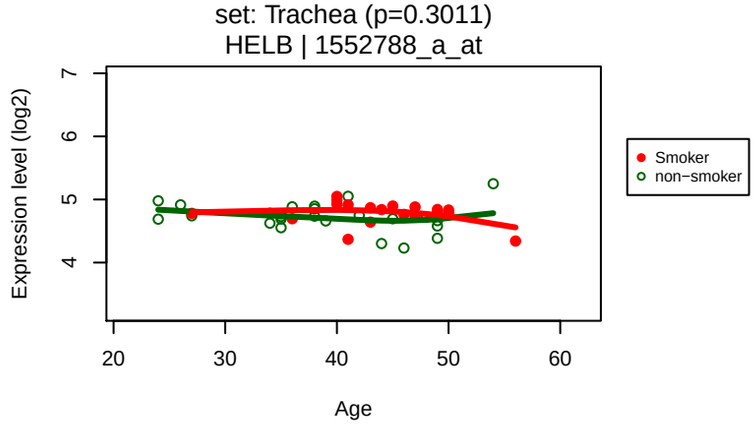

Gene expression ratio in Males
gene: HELB | 1552788_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.857
p=0.101
p=0.1144
p=0.2016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HELB | 1552788_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7595
p=0.4988
p=0.7508
p=0.4999
SmallAirway2
LargeAirway2
Trachea
Macrophage2

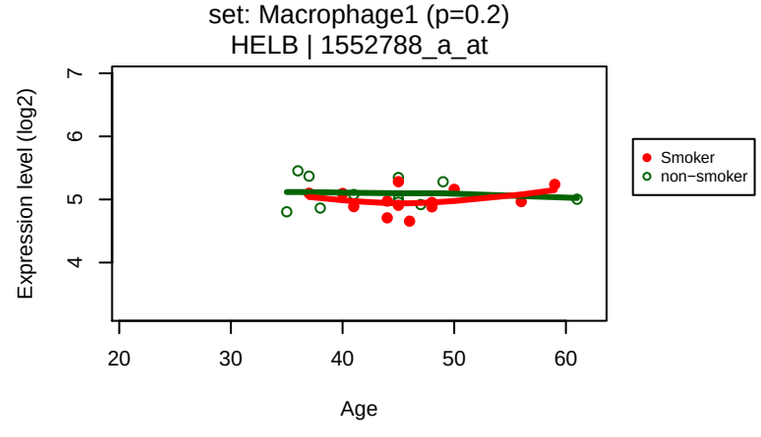

set: SmallAirway1 (p=0.2283)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker

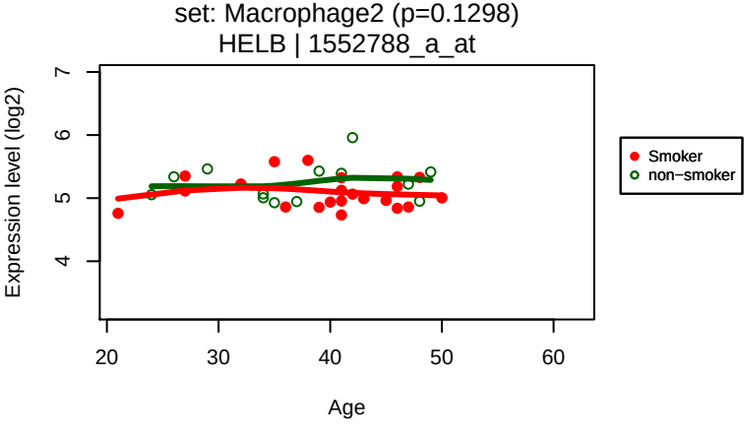

set: SmallAirway2 (p=0.8593)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker

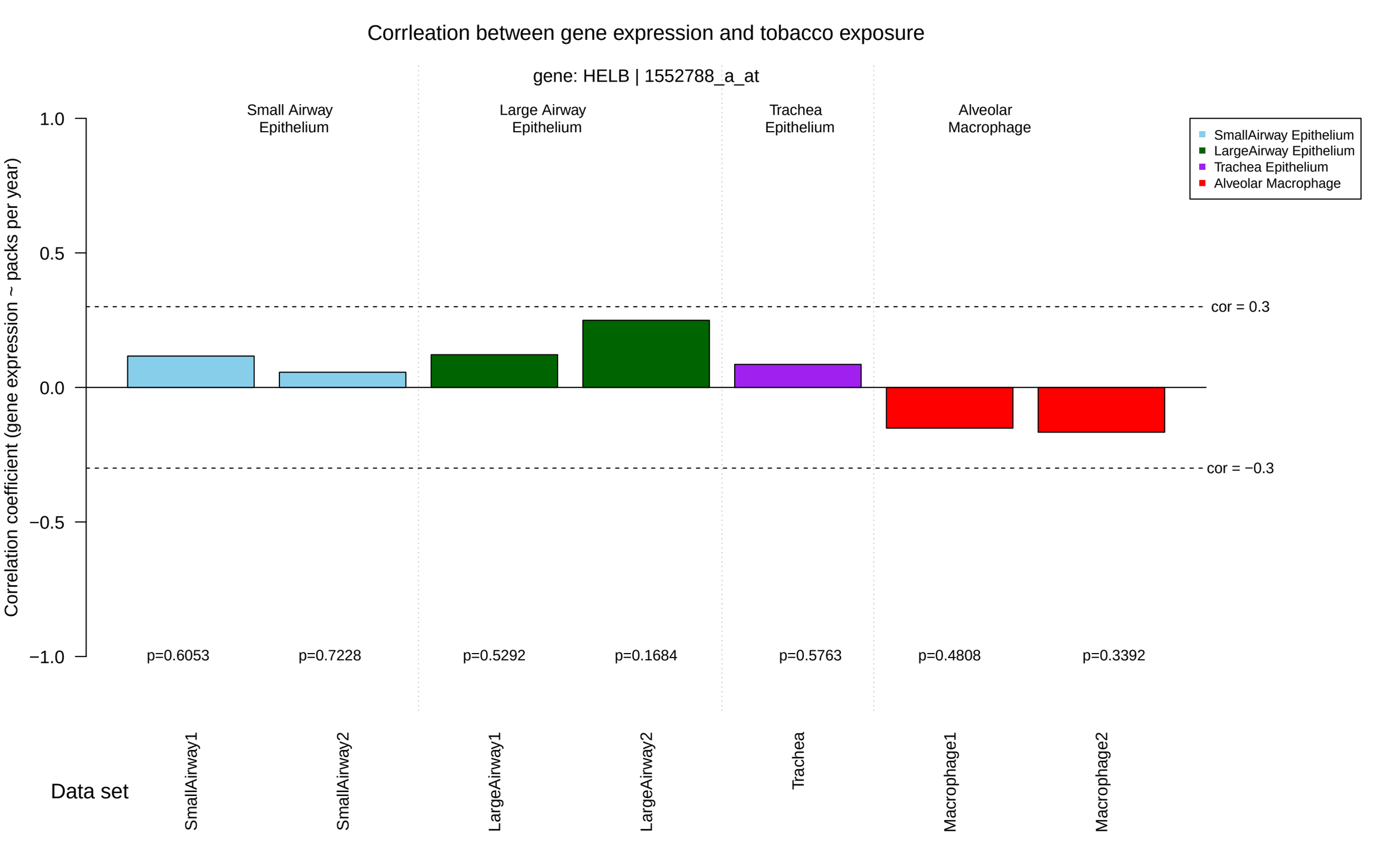

set: LargeAirway1 (p=0.0943)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0441)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3011)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1298)
HELB | 1552788_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HELB | 1552788_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6053
p=0.7228
p=0.5292
p=0.1684
p=0.5763
p=0.4808
p=0.3392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2